DISTANCE EDUCATION

M.Phil. (Mathematics) DEGREE EXAMINATION, MAY 2012.

COMMUTATIVE ALGEBRA

(Upto 2006 batch)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Define free module and prove that any two basis of a free module have the same cardinality. (10)

(b) Let $O \to N \to P \to M \to O$ and $O \to N' \to P' \to M \to O$ be two exact sequences of R – modules with p, p' projectives. Then prove that $P \oplus N' \sim P' \oplus N$. (10)

2. (a) Define Tensor product and prove that $M \otimes N \simeq \oplus \sum_{i,j} (M_1 \otimes N_1)$ where $M = \oplus \sum_{i=1}^{n} M_i$ and $N = \oplus \sum_{j=1}^{m} N_j$. (10)

(b) Define finitely generated and show that $A \underset{R}{\otimes} B$ is finitely generated R - algebra where A and B are finitely generated R – algebras. (10)

3. (a) (i) Define flat module with an example. (5)

(ii) Prove that M is faithfully flat if and only if M is flat and for each maximal ideal M of R, $MM \neq M$. (5)

(b) (i) Prove that the set of all Nilpotent elements of R is an ideal of R. (5)

(ii) Define Jacolyon radical of R with an example. (5)

4. (a) (i) Define Artimian ring with an example. (5)

(ii) Prove that if $\{M_i\}_{1 \leq i \leq n}$ are artisian R – modules, then $\oplus \sum_i M_i$ is artisian. (5)

(b) Let M be a finitely generated module over a Noetherian ring R. Then prove that the following are equivalent.

(i) $L_R(M)$ is finite.

(ii) Every $P \in$ Ass (M) is a maximal ideal.

(iii) Every $P \in$ Ass (M) is a maximal ideal. (10)

5. (a) Prove that if R is a Noetherian ring then the Nil radical $N(R) = \cap P$ PE ASS (R). (10)

(b) Prove that if M and N are projective R – modules of rank one, $M \underset{R}{\oplus} N$ and M^* are projective R – modules of rank one. (10)

6. (a) Prove that an R – module M has a composite series if and only if it is both a Noetherian and an Artinian module. (10)

(b) (i) State and prove Jordan – Holder theorem. (7)

(ii) Explain Discrete valuation rings. (3)

7. (a) State and prove Going-up theorem. (10)

(b) State and prove Norther's Normalization theorem. (10)

8. (a) Let R be a Dedekind domain. Let I be a non-zero Ideal of R. Them prove that every ideal of R/I is principal. (10)

(b) Prove that any R – module M can be embedded in an injective R – module. (10)

DISTANCE EDUCATION

M.Phil. (Mathematics) DEGREE EXAMINATION, MAY 2012.

MEASURE THEORY

(Upto 2006 batch)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. (a) Let u and v be real measurable functions on a measurable space X. Let ϕ be a continuous mapping of the plane into a topological space Y and define $h(x) = \phi(u(x), v(x))$ for $x \in X$ then prove that $h : X \to Y$ is measurable. (10)

(b) (i) Define measurable. (2)

(ii) Let $f : X \to [0, \infty]$ be measurable. There exists simple measurable functions S_n on X such that (1) $0 \le S_1 \le S_2 \le \ldots \le f$ and (2) $S_n(x) \to f(x)$ as $n \to \infty$ for every $x \in X$. (8)

2. (a) If $f_n : X \to [0, \infty]$ is measurable for $n = 1,2,3,..$ and $f(x) = \sum_{n=1}^{\infty} f_n(x)$ $(x \in X)$, then prove that $\int_X f d\mu = \sum_{n=1}^{\infty} \int_X f_n d\mu$. (10)

(b) Suppose f and $g \in L'(\mu)$ and α and β are complex numbers. Then prove that, $\alpha f + \beta g \in L'(\mu)$ and $\int_x (\alpha f + \beta f) d\mu = \int_x f d\mu + \beta \int_x g d\mu$. (10)

3. (a) Suppose $\{f_n\}$ is a sequence of complex measurable functions defined almost everywhere on X such that $\sum_{n=1}^{\infty} \int_x |f_n| d\mu < \infty$. Then prove that the series $f(x) = \sum_{n=1}^{\infty} f_n(x)$ converges for almost all $x, f \in L'(\mu)$ and $\int_x f d\mu + \sum_{n=1}^{\infty} \int_x f_n d\mu$. (10)

(b) (i) Let X and Y be topological spaces. Let $f : X \to Y$ be continuous. Prove that if K is a compact subset of X, then $f(K)$ is compact. (8)

(ii) Define support of a complex function. (2)

4. (a) Suppose X is a locally compact Hausdorff space, V is open in X, $K \subset V$ and K is compact. Then prove that there exist an $f \in C_c(X)$ such that $K < f < V$. (10)

(b) Let X be a locally compact Hausdorff space in which every open set is σ-compact. Let λ be any positive Borel measure on X such that $\lambda(K) < \infty$ for every compact set K. Then prove that λ is regular. (10)

5. (a) State and prove LUSIN'S theorem. (10)

(b) State and prove the Vitali-Caratheodory theorem.(10)

6. (a) State and prove Holder's and Minkowski's inequality. (12)

(b) Prove that if X is a locally compact Hausdorff space, then $C_o(x)$ is the completion of $C_c(x)$ relative to the metric defined by the supremum norm $\|f\| = \sup|f(x)| x \in X$. (8)

7. (a) (i) If $Z_1, Z_2 ... Z_N$ are complex numbers then prove that there is a subset S of $\{1,2,...N\}$ for which

$$\left|\sum_{K \in S} Z_K\right| \geq \frac{1}{n}\sum_{K=1}^{N}|Z_K|.$$ (8)

(ii) Define positive and negative variations. (2)

(b) Let $(x, \Im, \mu)$ and (y, $\mathscr{F}$, λ) be a finite space. Let f be an ($\Im \times \mathscr{F}$) measurable function on X × Y. Prove that,

(i) If $0 \leq f \leq \infty$ and if $\phi(x) = \int_y fxd\lambda$, $\psi(y) = \int_x f^y d\mu$ $(x \in X,\ y \in Y)$ then ϕ is $\Im$-measurable and ψ is $\mathscr{F}$–measurable. Also prove that $\int_X \phi d\mu = \int_{X \times Y} fd(\mu \times \lambda) = \int_Y \psi d\lambda$.

(ii) If F is complex and if $\phi^*(x) = \int_y |f|_x d\lambda$ and $\int_y \phi^* d\mu < \infty$, then $f \in L'(\mu \times \lambda)$. (10)

8. (a) Let M_k denote Lebesgue measure on R^K. If $K = r + s, r \geq 1, s \geq 1$ then prove that m_k is the completion of the product measure $m_r \times m_s$. (10)

(b) (i) Define convolution. (2)

(ii) Prove that if $E \in S \times Y$ then $E_x EJ$ and $E_y \in S$ for every $x \in X$ and $y \in Y$. (8)

DE–4291 **13**

DISTANCE EDUCATION

M.Phil. (Mathematics) DEGREE EXAMINATION, MAY 2012.

TOPOLOGICAL VECTOR SPACES

(Upto 2006 batch)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that, In a topological vector space X,

   (i) Every neighbourhood of O contains a balanced neighbourhood of O.

   (ii) Every convex neighbourhood of O contains a balanced convex neighbourhood of O. (10)

   (b) Let $\wedge$ be a linear functional on a topological vector space X. Assume $Ax \neq 0$ for some $x \in X$. Then prove that each of the following four properties implies the other three.

   (i) $\wedge$ is continuous.

   (ii) The null space $N(A)$ is closed

   (iii) $N(A)$ is not douse in X.

   (iv) $\wedge$ is bounded in some neighbourhood V of O. (10)

2. (a) If x is a topological vector space with a countable local base, then there is a metric d on X such that

(i) d is compatible with the topology of x,

(ii) the open balls centered at O are balanced and

(iii) d is invariant : $d(x+z,\ y+z)=d(x,y)$ for $x,y,z \in x$.

If, in addition, x is locally convex, then d can be chosen so as to satisfy (i), (ii) (iii) and also

(iv) all open balls are convex. (10)

(b) Prove that the following two properties of a sec E in a topological vector space are equivalent

(i) E is bounded

(ii) If $\{x_n\}$ is a sequence in E and $\{\alpha_n\}$ is a sequence of scalars such that $\alpha_n \to 0$ as $n \to \infty$, then $\alpha_n x_n \to 0$ as $n \to \infty$. (10)

3. (a) Show that, suppose A is a convex absorbing set in a vector space x, then

(i) $\mu_A(x+y) \le \mu_A(x)+\mu_A(y)$

(ii) $\mu_A(tx)=t\mu_A(x)$ if $t \ge 0$

(iii) μ_A is a seminorm if A is balanced

(iv) If $B=\{x : \mu_A(x)<1\}$ and $C=\{x:\mu_A(x)\le\}$, then $B \subset A \subset C$ and $\mu_B=\mu_A=\mu_C$. (10)

(b) State and prove the category theorem. (10)

4. (a) If f is a continuous linear functional on a subspace M of a locally convex space x, then prove that there exists $\wedge \in x^*$ such that $\wedge = f$ on M. (10)

(b) Suppose :

(i) x is an F space

(ii) y is a topological vector space

(iii) $A : x \to y$ is continuous and linear and

(iv) $A(x)$ is of the second category in y then

(1) $A(x) = y$

(2) A is an open mapping, and

(3) y is an $F-$space. (10)

5. (a) Suppose $B: X+Y \to Z$ is bilinear and separately continuous, x is an $F-$space, and y and z are topological vector spaces, then

$B(x_n, y_n) \to B(x_0, y_0)$ in z whenever $x_n \to x_0$ in X and $y_n \to y_o$ in Y. If Y is metrizable, it follows that B is continuous. (12)

(b) (i) Define dual space. (2)

(ii) Prove that, suppose A and B are disjoint, nonempty, convex sets in a topological vector space X.

(1) If A is open there exist $A \in X^*$ and $\nu \in R$ that $\text{Re} \wedge x < \nu \leq \text{Re} \wedge y$ for every $x \in A$ and for every $y \in B$.

(2) If A is compact, B is closed, and X is locally convex, then there exist $\wedge \in X^*$, $\nu_1 \in R$, $\nu_2 \in R$, such that $\mathrm{Re}\wedge x < \nu_1 < \nu_2 < \mathrm{Re}\wedge y$ for every $x \in A$ and for every $y \in B$. (8)

6. (a) Prove that X is a vector space and X' is a separating vector space of linear, functionals on X. Then the X'-topology T makes X into a locally convex space whose dual space in X'. (10)

(b) Prove that if V is a neighbourhood of O in a separable topological vector space X, and if $\{\wedge_n\}$ is a sequence in X^* such that

$$\left|\wedge_n x\right| \le 1 \quad (xeV_1\, n = 1, 2, 3, \ldots)$$

Then there is a sequence in X^* such that $\wedge_x = \lim\limits_{1\to\infty} \wedge_{ni} x (x \in X)$. (10)

7. (a) Prove that the following conclusion hold by assuming $f : \Omega \to X$ is weakly holomorphic.

(i) f is strongly continuous in Ω.

(ii) The Cauchy theorem and the Cauchy formula hold : If T is a closed path in Ω such that $ind_r(w) = 0$ for every $w \not\subset \Omega$, then

(1) $\int_T f(T_0)\, dT_0 = 0$, and

(2) $f(z) = \dfrac{1}{2zi} \int_T (T_0 - z)^{-1} f(T_0)\, dT_0$

If $z \in \Omega$ and $Ind_r(z) = 1$, If T_1 and T_2 are closed path in Ω such that $Ind_{r_1}(w) = Ind_{r_2}(w)$ for every $w \not\subset \Omega$, then

(3) $\int_{T_1} f(T_0)\, dT_0 = \int_{T_2} f(T_0)\, dT_0$.

(iii) f is strongly holomorphic in Ω. (15)

(b) Prove that suppose B is the closed unit ball of a normed space X. Define

$\|x^*\| = \sup\left\{\left|\langle x, x^*\rangle\right| : x \in B\right\}$ for every $x^* \in X^*$

(i) This norm makes X^* into a Banach space.

(ii) Let B^* be the closed unit ball of X^*. For every $x \in X$,

$\|x\| = \sup\left\{\left|\langle x, x^*\rangle\right| : x^* \in B^*\right\}$, consequently, $x^* \to \langle x, x^*\rangle$ is a bounded linear functional on X^*, of norm $\|x\|$.

(iii) B^* is weak* compact. (5)

8. (a) Prove that, if X is a Banach space, $T \in B(X)$, T is compact, and $\lambda \neq 0$, then $T - \lambda I$ has closed range. (10)

(b) Prove that, on every compact group G exists a unique a Borel probability measure m which is left-invariant in the sense that

(i) $\int_G f\, d_m = \int_G (L_s f)\, d_m [S \in G, f \in C(G)]$

This m is also right - invariant

(ii) $\int_G f d_m = \int (R_s f) d_m \; [S \in G, f \in C(G)]$ and it satisfies the relation.

(iii) $\int_G f(x)\, d_m(x) = \int_G f(x^{-1}) d_m(x) \; [F \in C(G)]$. (10)

DE–4292 | **14 (A)**

DISTANCE EDUCATION

M.Phil. (Mathematics) DEGREE EXAMINATION, MAY 2012.

FUNDAMENTALS OF DOMINATIONS IN GRAPHS

(Upto 2006 Batch)

Time : Three hours | Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) For any strongly perfect graph G, prove that $\beta_0(G) = \Gamma(G) = IR(G)$

   (b) If a graph G has no isolated vertices, show that $\gamma(G) \le \beta_2(G)$.

2. (a) If a graph G has $\gamma(G) \ge 2$, prove that $m \le \frac{1}{2}\left(n - \gamma(G)\right)\left(n - \gamma(G) + 2\right)$.

   (b) For any graph G, prove that $\ominus(G) = \Gamma(G)$.

3. (a) Prove the following :

   (i) If G is a graph with $\gamma(G) \ge 2$, then $r(G) = \mu(G)$.

   (ii) If a graph G has $\gamma(G) = 2$, then $r(G) = n - \Delta(G) - 1$.

   (b) Prove that for any tree T with $n \ge 2$, there exists a vertex $v \in V$ such that $\gamma(T - V) = \gamma(T)$.

4. (a) If a graph $G \in C \vee R$ has order $n = (\Delta(G)+1)(\gamma(G)-1)+1$, prove that G is regular.

(b) Show that a graph $G \in U \in A$ if and only if V^- empty.

5. (a) If G is a connected graph with $n \geq 3$ vertices, then prove that $\gamma_t(G) \leq \frac{2n}{3}$.

(b) For any connected graph G, show that diam $(G) - 1 \leq \gamma_c(G) \leq 2\beta_1(G)$.

6. (a) If a connected graph G has $n \geq 6$ and $\delta(G) \geq 2$, that prove that $\gamma_{pr}(G) \leq \frac{2n}{3}$.

(b) If a graph G does not contain the star $K_{1,k+1}, k \geq 2$, as an induced subgraph, that prove that $i(G) \leq (k-1)\gamma(G) - (k-2)$.

7. (a) What are the varieties of domination? What are the conditions imposed on each?

(b) If every vertex V in a tree T has odd degree, then prove that $PD(T) = 1$.

8. (a) Prove the following :

(i) If D is a γ-set of a graph G, then atleast one vertex in $V - D$ is dominated by no more than two vertices in D.

(ii) For any graph G, $\gamma_2(G) \leq \beta_2(G)$.

(b) If D is a symmetric digraph, prove that D has a kernel and $S \subseteq V$ is a kernel if and only if S is a maximal independent set.

DE–4293 **14 (B)**

DISTANCE EDUCATION

M.Phil. (Mathematics) DEGREE EXAMINATION,
MAY 2012.

DATA STRUCTURES AND ALGORITHMS

(Upto 2006 Batch)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carry 20 marks.

$(5 \times 20 = 100)$

1. Explain the concept of function prototype and function overloading in C++ by giving examples.

2. Explain the use of the following in an object oriented programming.

   (a) Inline function

   (b) Memory management operators

   (c) Relational and logical expressions

   (d) Virtual functions.

3. Write a C++ algorithm to convert on infix expression into prefix form using stacks and explain with an example.

4. Explain the features of a stack and a linear linked list and write a C++ class to implement push, pop and search operations of a stock using linked list.

5. Write a C++ algorithm to implement a Queue using arrays.

6. Write an algorithm and a C++ routine to perform each of the following operations

   (a) Append an element to the end of a list

   (b) Delete the nth element from a list

   (c) Return the sum of the integers in a list.

7. Write the algorithm to implement the in order, post order, and preorder tree traversals.

8. Explain the algorithms used to find the number of paths of a given length and number of total paths existing between two particular nodes in a graph.